DBSCAN: Density-Based Spatial Clustering of Applications with Noise

Tuesday, October 14, 2025 11:12 PM

Cluster = Dense region of points

Clusters are seperated by regions of low density.

↳ Classification depends on two parameters

→ ① Epsilon (ε): Radius → Defines neighborhood of each pt

→ ② MinPts : Min no. of pts requireed to form a dense region

The three types of points :

① CORE POINT : Has atleast MinPts (including itsel

② BORDER POINT : Is within the ε of a core point, but i Core point itself.

③ NOISE POINT : Not a core pt nor a border pt.

STRENGTHS

→ Can find clusters of arbitrary shape

→ Identifies Outliers ✓

→ No need to specify k

WEAKNESS :

→ Parameter Sensitivity : Very imp to choose correct ε

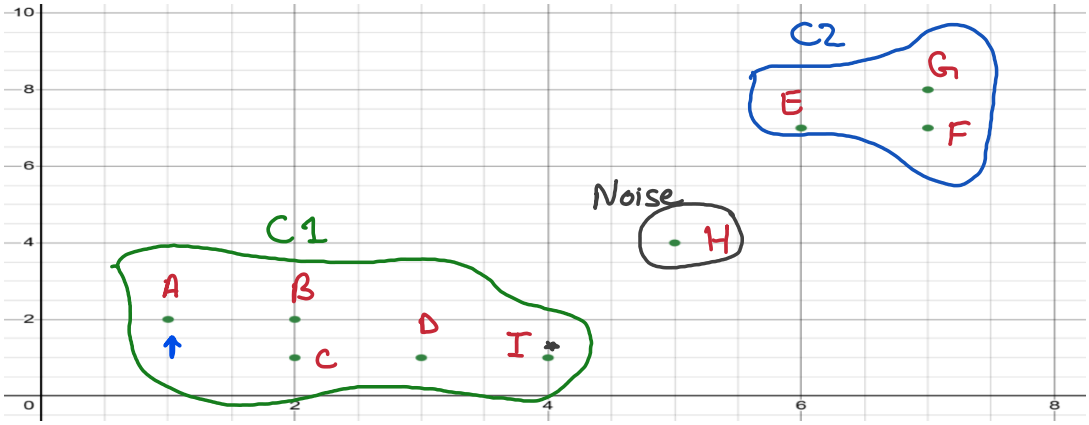
Example

Dataset : 8 points (A to H)

$\varepsilon = 1.5$

MinPts = 3

A = (1,2) E = (6,7) I = (4,1)
B = (2,2) F = (7,7)
C = (2,1) G = (7,8)
D = (3,1) H = (5,4)



Distance:

→ Pts that are 1 unit apart vertically or horizontally = 1
→ Pts that are 1 unit apart diagonally = $\sqrt{1^2+1^2} \simeq 1.41$

→ Anything else is > 1.5

Initial Pass

A → B(1), C(1.41) → Core Pt

Core?

B → A, C, D → Yes 4 ≥ 3 ✓
C → A, B, D → Yes
D → B, C, I → Yes
E → F, G → Yes
F → E, G → Yes 3 ≥ 3 ✓
G → E, F → Yes
H → None → No

I → D → No ? ≥ 3 ×

Grow the clusters:

① Pick A
- It is a core pt → we create cluster 1 → Add A to C1 {A}
- Add neighbors of A to C1 → {A, B, C}

② Expand next core pt → Expand B
- Add neighbors to C1 → {A, B, C, D}

③ Expand C → {A, B, C, D}

④ Expand D → Neighbors = B, C, I → Add I to C1 {A, B, C, D, I}

Is I a core pt? → No → Border Point!
↳ Do not expand I

⑤ Move to next unvisited pt → E → Core → Add to C2
Add neighbors to C2 → {E, F, G}

⑥ Expand F and G → {E, F, G} = C2

⑦ Move to last point → H
Core? → No ×
Classify as Noise ✓

C1 = {A, B, C, D, I*}

C2 = {E, F, G}

Noise = {H}